

Element E

Our design for the rose is a cup with a hole in the bottom that is attached to a pipe that is then on a circular wooden base. The petals will be attached to paperclips on the outside of the cup. The paperclips are then attached to a fishing line that is then put through the pipe and through the bottom of the table. When the strings are pulled they will go through small holes in the cup which then make the petals dislodge from the paperclip and fall off of the rose. During our planning and designing we got the chance to speak to 2 experts who helped us make our vision into a tactical mechanism. One was an engineering PHD candidate at Vanderbilt university, and the other was a theater professor at Vanderbilt. We showed them our design and got many helpful comments and ideas. We showcased the STEM principles of our design by hooking it up and demonstrating one of the string mechanisms of our prototype, which successfully dropped the petal to the ground. Our engineering advisor gave us some positive feedback when the rose petal dropped, and gave the following suggestions to improve the viability of the project. She said to prevent the petals from falling off prematurely we could tape the backs with clear/scotch tape to keep them attached to the hook before it is pulled. She also said that tying the strings together and pulling a singular string with multiple petals connected to it would be a good way to have all of the petals fall off in a timely manner. Advice from our theater production expert was to make sure that the person who is controlling the petals knows how many petals are supposed to fall off at a time to enhance the magic. He also advised to make all of the details of the paint job and petal shape very pronounced so that people in the back of the theater can see exactly what is going on with the rose on stage. All in all, our rose design showcases STEM principles such as the angle of the paper clips attached to the petals to make sure they don't fall off prematurely, and our selection of the best materials to make the rose functional and cost effective.

